

Chapter 1 Introduction: An Overview of Some Topics

Chapter 2 Lie Groups

2.1 Lie Groups

2.1.1 Definition

Definition 2.1.1.1 A **Lie group** is a group G , endowed with the structure of smooth manifold, such that the operations

$$\begin{aligned} m_G : G \times G &\rightarrow G & (x, y) &\mapsto xy \\ \iota_G : G &\rightarrow G & x &\mapsto x^{-1} \end{aligned}$$

are smooth.

In addition,

Definition 2.1.1.2 Equipped with the maps

$$\begin{aligned} \lambda_g : G &\rightarrow G, x \mapsto gx \\ \rho_g : G &\rightarrow G, x \mapsto xg \\ c_g : G &\rightarrow G, x \mapsto gxg^{-1} \end{aligned}$$

They are bijective with

$$\begin{cases} \lambda_{g^{-1}} = (\lambda_g)^{-1} \\ \rho_{g^{-1}} = (\rho_g)^{-1} \\ c_{g^{-1}} = (c_g)^{-1} \end{cases}$$

$(c_g)_{g \in G}$ is a homomorphism of groups, with

$$c_{(-)} : G \hookrightarrow \text{Aut}(G) \quad g \mapsto c_g$$

Its image is a subgroup of G , denoted by $\text{Inn}(G)$.

♣ **Example 2.1.1.3** Some are listed as followings:

- $(\mathbb{R}, 0, +, -)$: Non-compact, connected;
- $(\mathbb{R}, 1, \cdot, (-)^{-1})$: Non-compact, non-connected;
- $(GL_N(\mathbb{R}), 1_N, \cdot, (-)^{-1})$. Non-compact, non-connected;
-
-

◇ **Lemma 2.1.1.4** ...

◦ **Proof** ...

□

★ **Corollary 2.1.1.5** ...

Morphisms of Lie Groups

Definition 2.1.1.6 The morphism of Lie groups is a smooth homomorphism

$$\varphi : G_1 \rightarrow G_2$$

Lie Subgroups

2.1.2 The Identity Component

2.1.3 Invariant of Vector Fields

2.1.4 Classification of Abelian Lie Groups

2.2 Covering of Lie Groups

2.3 Matrix Groups

Some Works with Determinant

◇ **Lemma 2.3.0.1** Determinant

$$\det : \text{Mat}_n(F) \rightarrow F$$

is continuous on $\text{Mat}_n(F)$.

◦ **Proof** Notice that

$$\begin{aligned} \|\det(A + H) - \det(A)\| &= \|\det(A)[\det(1 + A^{-1}H) - \det(1)]\| \\ &= \|\det(A)(\text{tr}(A^{-1}H))\| = O(\|H\|) \end{aligned}$$

□

◇ **Lemma 2.3.0.2** The map $\det : \text{Mat}_n(\mathbb{R}) \rightarrow \mathbb{R}$ is differentiable at any $A \in GL_n(\mathbb{R})$, and we have

$$d[\det](A) : \text{Mat}_n(\mathbb{R}) \rightarrow \mathbb{R} \quad H \mapsto \det(A)\text{tr}(A^{-1}H)$$

◦ **Proof** Firstly, in the case of $A = 1$:

$$d[\det](1)H = (\det(1 + H) - \det(1)) \cdot 1 + o(\|H\|)$$

And, let $H = [h_1 | \dots | h_n]$, we use the following inequality

$$O(\|H\|^n) \leq O(\|H\|^2) \leq o(\|H\|)$$

$$\begin{aligned} \det(1 + H) &= \det([e_1 + h_1 | \dots | e_n + h_n]) \\ &= \text{tr}(H) + \det([e_1 | \dots | e_n]) + o(\|H\|) \\ &= 1 + \text{tr}(H) + o(\|H\|) \\ \Rightarrow d[\det](1)H &= \text{tr}(H) + o(\|H\|) \end{aligned}$$

In general:

$$\begin{aligned} d[\det](A)H &= (\det(A + H) - \det(A)) + o(\|H\|) \\ &= \det(A)(\det(1 + A^{-1}H) - \det(1)) + o(\|H\|) \\ &= \det(A) \cdot d[\det](1)(A^{-1}H) = \det(A)\text{tr}(A^{-1}H) \end{aligned}$$

□

★ **Corollary 2.3.0.3** 1 is the regular value of $\det$.

◦ **Proof** We have: 1 is the regular value of $\det \Leftrightarrow \det(A) = 1 \rightarrow d[\det](A) \neq 0$. Thus, action on $X = A$,

$$\det(A) = 1 \Rightarrow d[\det](A)A = \det(A)\text{tr}(A^{-1}A) = n1$$

□

We can combine this lemma with **regular value's theorem**,

2.3.1 Orthonormal Groups

Definition 2.3.1.1 Define:

$$O(n; \mathbb{R}) := \{X \in M_n(\mathbb{R}) \mid X^T = -X\}$$

Especially,

Definition 2.3.1.2 Let V be a finite-dimensional vector space, with $\langle -, - \rangle$ a non-degenerate symmetric bilinear form on V (not assume positive definite). Define:

$$O(\langle -, - \rangle; \mathbb{R}) := \{A \in GL(V) \mid \langle Ax, Ay \rangle = \langle x, y \rangle; x, y \in V\}$$

It is a disconnected group.

So in $\mathbb{R}^n$, the inner product $\langle x, y \rangle := x^T y$

◇ **Lemma 2.3.1.3** $O(\langle -, - \rangle)$ is disconnected.

◦ **Proof** Define: $\det(-) : GL(V) \rightarrow \mathbb{R}$, then we have $\det(A) \neq 0, A \in GL(V)$.

□

◇ **Lemma 2.3.1.4**

- The map

$$\Phi : \text{End}(V) \rightarrow S^2 V^* \quad A \mapsto A^* \langle - | - \rangle$$

is differentiable at $A \in \text{End}(V)$.

- For $A \in O(\langle -, - \rangle)$ we have

$$d\Phi_A : \text{End}(V) \rightarrow S^2 V^* \quad H \mapsto (\mapsto)$$

◦ **Proof ...**

□

★ **Corollary 2.3.1.5** ...

◦ **Proof ...**

□

Affine Linear Transformations

Minkowski 4-Space

Isometry Group of Sphere

2.4 Exponential Function

2.4.1 The Matrix Exponential Map

Definition 2.4.1.1 Define the matrix exponential of matrix: For a complete field

$$\exp(X) := \sum_{k=0}^{\infty} \frac{1}{k!} X^k$$

♠ **Proposition 2.4.1.2** ...

◦ **Proof** ...

□

2.4.2 The Exponential Map

♠ **Proposition 2.4.2.1** Let $X \in \mathfrak{g}$. There exists a **unique group homomorphism**

$$\gamma_X : \mathbb{R} \rightarrow G \quad \gamma'_X(0) = X$$

◦ **Proof**

- Existence:
- Uniqueness:

□

Definition 2.4.2.2 For such γ_X , define:

$$\exp_G(X) := \gamma_X(1)$$

► **Theorem** The map $\exp_G : \mathfrak{g} \rightarrow G$ is smooth, with following properties:

- (Commutativity) $Ad(x) \circ \exp_G = \exp_G \circ Ad(x); x \in G;$
- $Ad \circ \exp_G = \exp_{GL(\mathfrak{g})} \circ ad;$
- $d\exp_G(0) = id_{\mathfrak{g}};$
- If $\phi : G \rightarrow H$ is a homomorphism, then

$$\phi \circ \exp_G = \exp_H \circ d\phi(1)$$

•

◦ **Proof ...**

□

♠ **Proposition 2.4.2.3** (Commutativity of Exponential Maps) Let G be a Lie Group, with $X, Y \in \mathfrak{g}$.
If $[X, Y] = 0$, then

$$\exp_G(X) \cdot \exp_G(Y) = \exp_G(Y) \cdot \exp_G(X)$$

◦ **Proof ...**

□

2.4.3 Application: Homomorphism of Lie Groups on Connected Groups

◇ **Lemma 2.4.3.1** Let $f_1, f_2 : H \rightarrow G$ be homomorphism of Lie groups. If H is connected, and $df_1(1) = df_2(1)$, then

$$f_1 = f_2$$

◦ **Proof ...**

□

2.5 Linear Lie Group

2.6 Representation of Lie Groups

Definition 2.6.0.1 ...

► **Theorem ...**

Review of Some Linear Algebra

Review of Functional Analysis

2.6.1 Haar Integral

2.6.2 Representation

Examples: Representation of $SU(2)$, $SO(3)$, $U(2)$, $O(3)$

Chapter 3 Lie Algebras

3.1 Elementary Theories

3.1.1 Basic Concepts

Lie Algebras

Definition 3.1.1.1 A **Lie algebra** is a finite-dimensional real/complex vector space, together with a bilinear map

$$[-|-] : L \times L \rightarrow L$$

with following conditions:

-
-

♠ **Proposition 3.1.1.2** ...

◦ **Proof** ...

□

Homomorphisms

3.2 Root Decompositions

3.3 Representation Theory of Lie Algebra

Chapter 4 Basic Lie Theory

4.1 Lie Groups and their Lie Algebras

4.1.1 Lie Groups and Tangent Groups

In addition, c_g is a homomorphism of groups

$$c_- : G \hookrightarrow \text{Aut}(G) \quad g \mapsto c_g$$

And the image can be denoted as $\text{Inn}(G)$.

♣ **Example 4.1.1.1** Consider the following additive group $G = (\mathbb{R}^n, +)$, endowed with the n -dimensional manifold structure. Chart

$$(\mathbb{R}^n, id_{\mathbb{R}^n})$$

Product of groups: Induced by Cartesian Product structure.

♣ **Example 4.1.1.2** ...

4.1.2 Tangential Map

Similarly,

$$d(\lambda_g)_h : T_h G \rightarrow T_{gh} G$$

$$d(\rho_g)_h : T_g G \rightarrow T_{gh} G$$

4.1.3 Lie Group Structure of $T(G)$

◇ **Lemma 4.1.3.1** Let G be a Lie group:

- Tangent map

$$\begin{aligned} T(e_G) : T(G \times G) &= T(G) \times T(G) \rightarrow T(G) \\ (X_g, Y_h) &\mapsto d(\lambda_g)_h Y_h + d(\lambda_h)_g X_g \end{aligned}$$

defines a Lie-group structure on $T(G)$. And $T(G)$ has:

- Zero $0_1 \in T_1(G)$;
- Inverse $T(\iota_G)$;

And $\pi_{T(G)} : T(G) \rightarrow G$ is a morphism of Lie groups. With kernel

$$\ker(\pi_{T(G)}) = T_1(G)$$

Zero section $\sigma : G \rightarrow T(G), g \mapsto 0_g$ is a homomorphism of Lie groups. And

$$\pi_{T(G)} \circ \sigma = id_G$$

- Map

$$\Phi : G \times T_1(G) \rightarrow T(G)$$

is a diffeomorphism.

◦ **Proof**

- Group axioms:
 - (Associativity) $m_G \circ (m_G \times id_G) = m_G \circ (id_G \times m_G) \Rightarrow T(m_G) \circ (T(m_G) \times T(id_G)) = T(m_G) \circ (T(id_G) \times T(m_G))$;
 - (Unit)
 - (Inversion)
-

□

Remark In vector bundle, we usually have $T(G \times G) = T(G) \oplus T(G)$. But for finite dimensional G , $T(G) \oplus T(G) \cong T(G) \times T(G)$, abelian category.

4.1.4 The Lie Functor

Definition 4.1.4.1 (The Lie algebra of G) A vector field $X \in \mathcal{V}(G)$ is **left invariant** if

$$X = T(\lambda_g) \circ X \circ \lambda_g^{-1}, \forall g \in G$$

Define: $(\lambda_g)_* X := T(\lambda_g) \circ X \circ \lambda_g^{-1}$.

Write $\mathcal{V}^l(G) := \{X \in \mathcal{V}(G) | X \text{ is invariant}\}$.

♠ **Proposition 4.1.4.2** ...

◦ **Proof** ...

□

Isomorphism $T_1(G) \cong \mathcal{V}(G)^l$

★ **Corollary 4.1.4.3** If G acts freely on M , then there exists an isomorphism

$$T_1(G) \cong \mathcal{V}(G)$$

◦ **Proof** ...

□

4.1.5 Smooth Actions of Lie Groups

Definition 4.1.5.1 ...

4.1.6 Topology of Lie Groups

4.2 Exponential Functions

4.2.1 Basic Properties of Exponential Function

Definition 4.2.1.1 ...

Chapter 5 Smooth Action of Lie Groups

Chapter 6 Structure Theory of Lie Groups

Chapter 7 Compact Lie Theory

7.1 Clasical Groups

7.2 Vector Invariants

7.2.1 The Main Proposition of the Theory of Invariants

Definition 7.2.1.1

- Consider the following **elementary symmetric functions**: For vectors $x = (x_1, \dots, x_n)$, define

$$\begin{aligned}\varphi_0(x) &= 1 \\ \varphi_1(x) &= \sum_{i_1} x_{i_1} \\ \varphi_2(x) &= \sum_{i_1, i_2} x_{i_1} x_{i_2} \\ &\dots \\ \varphi_n(x) &= x_1 \dots x_n\end{aligned}$$

- Then, we have the polynomial coefficient for $\Phi(t)$:

$$\Phi(t) = (t - x_1) \dots (t - x_n) = \sum_0^n (-1)^i t^{n-i} \varphi_i(x)$$

- Furthermore, if $f(x_1, \dots, x_n)$ is any symmetric function of arguments $x_1, \dots, x_n$, there exists a function

This is the altest form of FMT.

Furthermore, we have the follwing statement the functions,

Claim If $f(x_1, \dots, x_n)$ is any symmetric function of n arguments $x_1, \dots, x_n$, there exists a function $F(\xi_1, \dots, \xi_n)$ of n arguments $\xi_1, \dots, \xi_n$ such that

$$f(x) = F(\varphi_1(x), \dots, \varphi_n(x))$$

Integral Rational Basis

Definition 7.2.1.2 ...

First Example: Orthogonal Transformation

Define: $\Gamma = O(n)$. Then, take $x, y \in F^n$,

...

Orthogonally Invariant Form